

RAMC Phase 3: Proposed Extensions

Summary of Discussion on Occupancy Uncertainty
and Plant–Model Mismatch

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Contents

1	Context: Limitations of the Current Experimental Design	2
2	Task 1: Persistent Occupancy Uncertainty Layer	2
2.1	Description	2
2.2	Implementation	2
2.3	What Changes in the Experimental Matrix	3
2.4	Expected Outcome	3
3	Task 2: Genuine Plant–Model Mismatch via EnergyPlus Training	3
3.1	Description	3
3.2	Why This Is More Realistic	4
3.3	What Changes	4
3.4	Expected Outcome	4
4	Task 3: Combined Design (Recommended)	5
4.1	Description	5
4.2	Expected Outcome	5
5	Implementation Priority and Effort Estimate	6

1 Context: Limitations of the Current Experimental Design

The current Phase 3 closed-loop experiments (60 experiments: 4 models $\times$ 3 scenarios $\times$ 5 seeds) provide clean evidence that RAMC’s CVaR-based training loss improves closed-loop thermal comfort under identical MPC conditions. However, two simplifications limit the realism of the evaluation and may understate the practical advantage of risk-aware model training.

Simplification 1: Perfect internal-gains forecast. The MPC receives the exact time-varying Q_{internal} profile at every timestep over its full 24-hour horizon. In reality, occupant-driven heat gains (people, equipment, lighting) are a significant and largely unpredictable disturbance. This simplification removes an entire channel of forecast–truth mismatch from the experiments.

Simplification 2: No plant–model mismatch. The neural network is trained on data generated by the RC model, and the same RC model serves as the closed-loop plant. The NN’s prediction errors are therefore pure function-approximation errors that could, in principle, be driven to zero with sufficient capacity and data. In real deployment, the NN would be trained on building data (or EnergyPlus data) while the physical building (or a different model of it) serves as the plant. The structural disagreement between these two representations is irreducible and permanent.

Observed pattern motivating extensions. The current results show a clear gradient: RAMC’s advantage over Fidelity is *smallest* under nominal conditions (perfect forecast) and *largest* under forecast mismatch (cold snap, forecast error). Both proposed extensions introduce additional forecast–truth mismatch, and based on the observed pattern, we expect them to widen the Fidelity–RAMC performance gap.

2 Task 1: Persistent Occupancy Uncertainty Layer

2.1 Description

Internal heat gains (Q_{internal}) represent a source of uncertainty that is *independent of the weather scenario*. Unlike outdoor temperature or solar radiation, occupancy uncertainty exists on nominal days, during cold snaps, and under forecast errors alike. It should therefore not be treated as a separate scenario but as a **persistent uncertainty layer applied across all scenarios**.

The proposed modification: in every experiment, the MPC continues to receive the original deterministic Q_{internal} forecast (the training-episode profile), but the RC plant receives a *perturbed* version. The perturbation is generated once per experiment using the seed (maintaining determinism and reproducibility) but is never revealed to the MPC.

2.2 Implementation

The perturbation would be applied in the simulator’s `_build_disturbance_truth` method. A multiplicative stochastic scaling factor is applied to the forecast Q_{internal} to generate the truth value seen by the plant:

$$Q_{\text{internal}}^{\text{truth}}(k) = \alpha(k) \cdot Q_{\text{internal}}^{\text{forecast}}(k) \quad (1)$$

where $\alpha(k)$ is a time-correlated random process with mean 1.0, generated from the experiment seed. A suitable model would be a log-normal AR(1) process:

$$\ln \alpha(k) = \rho_{\alpha} \ln \alpha(k-1) + \varepsilon(k), \quad \varepsilon(k) \sim \mathcal{N}(0, \sigma_{\alpha}^2(1 - \rho_{\alpha}^2)) \quad (2)$$

with suggested parameters $\rho_{\alpha} = 0.95$ (strong temporal correlation — occupancy doesn’t change every 10 minutes) and $\sigma_{\alpha} = 0.3$ (30% standard deviation in the scaling factor). This produces

realistic variation: some hours the building has 70% of expected occupancy, other hours 130%, with smooth transitions.

2.3 What Changes in the Experimental Matrix

The matrix structure remains identical: 4 models $\times$ 3 scenarios $\times$ 5 seeds = 60 experiments. No new scenarios are added. The only change is that every experiment now has an additional source of forecast–truth mismatch on the Q_{internal} channel. Critically, the perturbation realisation for a given seed is identical across all four models within each scenario, preserving the paired-comparison structure.

2.4 Expected Outcome

Wider Fidelity–RAMC gap, especially in the nominal scenario. The nominal scenario currently represents the “easy” condition where the MPC has perfect information on all disturbance channels. With Q_{internal} uncertainty, the nominal scenario becomes “perfect weather forecast but imperfect occupancy forecast,” which is much closer to real operation. The MPC will occasionally under-predict or over-predict internal gains, leading to temperature excursions that the controller must recover from. The model’s accuracy in the low-temperature region — where RAMC’s training objective concentrates prediction quality — becomes more important.

Compounding effect with weather mismatch. In the cold snap and forecast error scenarios, Q_{internal} uncertainty compounds with weather mismatch. The MPC is now wrong about *two* disturbance channels simultaneously. This creates more frequent and more severe tail violations, amplifying the regime where RAMC’s CVaR-shaped accuracy provides the greatest advantage.

Quantitative expectation. Based on the current gradient (RAMC’s degree-hour improvement grows from -21% under nominal to -18% under cold snap with weather mismatch alone), we expect Q_{internal} uncertainty to push the nominal-scenario improvement closer to the cold-snap level (-18% to -25%) and to increase the cold-snap improvement beyond -18% .

3 Task 2: Genuine Plant–Model Mismatch via EnergyPlus Training

3.1 Description

The current training pipeline is:

$$\text{EP data} \xrightarrow{\text{system ID}} \text{RC model} \xrightarrow{\text{simulation}} \text{RC data} \xrightarrow{\text{training}} \text{NN model}$$

The NN learns to approximate the RC, and the RC serves as the plant. No structural mismatch exists.

The proposed pipeline is:

$$\begin{aligned} \text{EP data} &\xrightarrow{\text{system ID}} \text{RC model} \quad (\text{used as plant}) \\ \text{EP data} &\xrightarrow{\text{training}} \text{NN model} \quad (\text{used in MPC}) \end{aligned}$$

Both the RC and the NN are independently fitted to the same EnergyPlus data. Neither perfectly reproduces EnergyPlus. Crucially, they fail in *different ways*: the RC has structural limitations (6 states, simplified heat transfer) while the NN has capacity limitations (finite training data, potential overfitting, extrapolation fragility). The disagreement between them is *irreducible* — no amount of NN training can eliminate it, because the NN’s training target (EnergyPlus) and the evaluation plant (RC) are structurally different models.

3.2 Why This Is More Realistic

In real deployment, the situation mirrors this proposed pipeline exactly:

- A building exists (analogous to EnergyPlus — the ground truth data source).
- An RC model is identified from building sensor data (analogous to fitting RC to EP data).
- An NN is trained from the same sensor data (analogous to fitting NN to EP data).
- The MPC uses the NN; the building is the plant.

The NN and the building disagree structurally, just as the NN and the RC would disagree in the proposed setup. The current setup, where the NN is trained on RC outputs and evaluated against the same RC, is artificially favourable because it eliminates this structural disagreement.

3.3 What Changes

NN training. All four neural networks (Raw MSE, Fidelity, RAMC $\lambda = 0.001$, RAMC $\lambda = 0.005$) must be retrained using EnergyPlus simulation data as the training target, rather than RC simulation data. The training episodes, architecture, and loss functions remain the same — only the data source changes.

Evaluation plant. The RC model continues to serve as the closed-loop plant, using the same identified parameters. No changes to the simulator infrastructure.

Nature of mismatch. The NN will learn EnergyPlus dynamics, which include phenomena the RC cannot represent: detailed convective heat transfer, moisture interactions, ground coupling, window-frame thermal bridging, etc. When the MPC plans using the NN, it may predict thermal responses (e.g., faster solar gain absorption, different radiator emission curves) that the RC plant does not exhibit. This creates a *systematic* prediction bias that varies with operating conditions — exactly the type of mismatch encountered in practice.

3.4 Expected Outcome

Higher baseline violation levels for all models. The structural mismatch will cause all four models to perform worse in absolute terms compared to the current RC-trained setup. The MPC’s predictions will be systematically off in ways that cannot be corrected by function-approximation improvements alone.

Greater differentiation between RAMC and Fidelity. The systematic mismatch will produce more frequent and more severe tail violations. Under the current setup, the NN approximates the RC well enough that most comfort violations arise from forecast mismatch or initial-condition effects. Under the proposed setup, model mismatch becomes a persistent additional source of violations. RAMC’s training objective — which concentrates prediction accuracy in the high-violation region — would provide a larger advantage because there are simply more tail events for the risk-aware shaping to mitigate.

More realistic assessment of deployment viability. If RAMC’s advantage holds (or grows) under genuine plant–model mismatch, it would demonstrate that risk-aware training is robust to the conditions encountered in real buildings. This would substantially strengthen the practical impact argument.

Potential risk. If the RC–EP mismatch is very large relative to the NN’s approximation error, all four models may perform poorly and the differences between them may be dwarfed by the structural mismatch. In this case, the conclusion would be that model architecture or plant identification matters more than the training loss — itself a useful finding, though not

the one the paper aims for. The likelihood of this outcome depends on the quality of the RC identification; given that the RC was carefully fitted with differential evolution and validated in Phase 2, the mismatch should be moderate rather than overwhelming.

4 Task 3: Combined Design (Recommended)

4.1 Description

The strongest experimental design combines both extensions: NN models trained on EnergyPlus data, evaluated against an RC plant, with Q_{internal} uncertainty applied as a persistent layer across all scenarios.

Component	Configuration
NN training data	EnergyPlus episodes
Evaluation plant	RC model (identified from EP data)
Weather forecast mismatch	Per scenario (nominal / cold snap / forecast error)
Q_{internal} mismatch	Persistent stochastic layer (all scenarios)
MPC controller	Identical across all models (unchanged from current)

This creates three simultaneous and independent sources of closed-loop difficulty:

1. **NN approximation error:** The NN does not perfectly reproduce EnergyPlus dynamics (training-data limitation).
2. **Plant-model structural mismatch:** The RC plant does not behave like EnergyPlus (model-class limitation).
3. **Disturbance forecast mismatch:** Weather scenarios plus Q_{internal} uncertainty (operational uncertainty).

Each source affects all four models, but RAMC’s CVaR-shaped training concentrates accuracy in the high-violation states that arise from all three sources. The combined design tests whether risk-aware model training provides robust benefit under realistic deployment conditions.

4.2 Expected Outcome

The combined design would produce the largest Fidelity–RAMC differentiation, because every additional source of uncertainty creates more tail events for RAMC’s risk-aware accuracy to mitigate. The experimental matrix structure ($4 \times 3 \times 5 = 60$ experiments) remains unchanged, preserving the paired-comparison methodology.

If the current results show a median CVaR₉₀ improvement of -0.20°C and a degree-hour reduction of -18% to -27% under artificially favourable conditions (no plant-model mismatch, perfect Q_{internal} forecast), the combined design would test whether these advantages grow under realistic conditions. A positive result would establish RAMC not just as a theoretically cleaner training paradigm, but as a practically deployable improvement for real building control systems.

5 Implementation Priority and Effort Estimate

Table 1: Implementation summary for proposed extensions.

Task	Effort	Impact	What Changes
Task 1: Q_{internal} uncertainty	Low	Moderate-High	Modify <code>_build_disturbance_truth</code> in simulator; re-run 60 experiments (same models, same matrix). No retraining.
Task 2: EP-trained NNs	High	High	Retrain all 4 NN models on EnergyPlus data; re-run 60 experiments. Training pipeline changes; simulator unchanged.
Task 3: Combined	High	Highest	Both Task 1 and Task 2 changes. Single re-run of 60 experiments captures both effects simultaneously.

Recommendation. If the current paper is close to submission, publish the existing clean results and frame Tasks 1–3 as future work with explicit testable predictions. If time permits before submission, Task 1 alone (low effort, no retraining) would meaningfully strengthen the practical impact argument. Task 2 and the combined Task 3 represent a natural follow-up study.